

Data description: Numerical and Graphical summaries (for Numerical variables)

Presentation Outline

- 1 Numerical Representations of Data
- 2 Graphical Representations of Data
- 3 Graphical Representations for Two or More Variables
- 4 Important points

Describing the Location (Typical Value)

Notation: Σ : sum, X_i : i -th observation

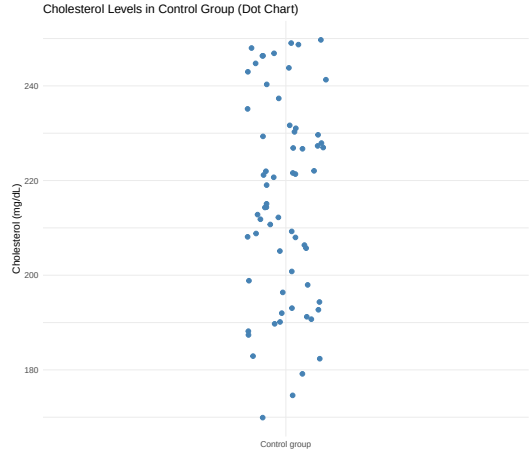
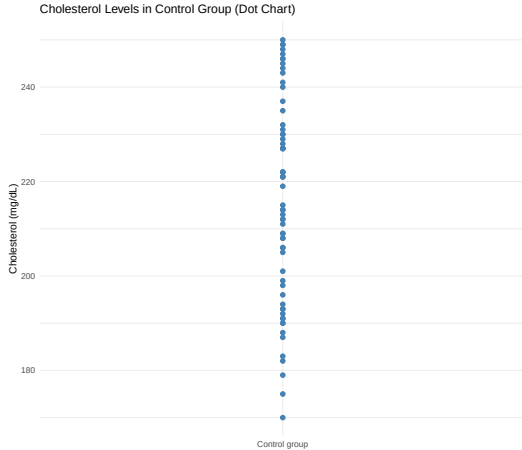
- **Mean:** $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$
- **Median:** value with 50% above and 50% below in the ordered data
- *Note:* The median is also the 2nd quartile (Q_2).
What are Q_1 and Q_3 ? $\rightarrow$ 25th and 75th percentiles

Describing the Spread (Variability)

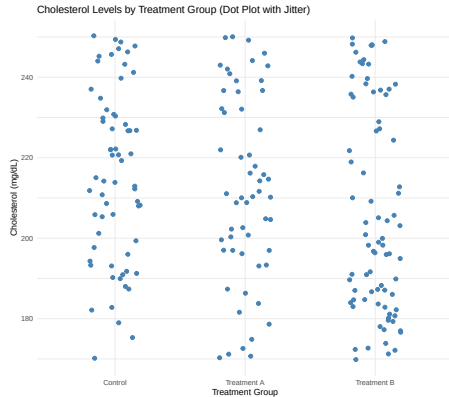
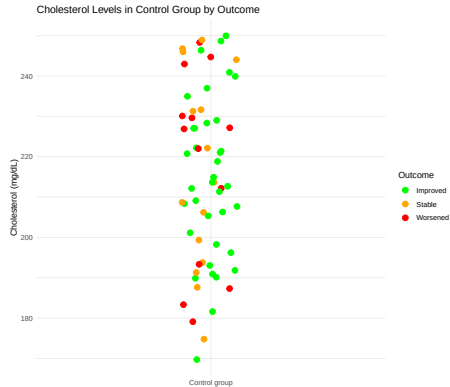
- **Variance:** $s^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$
- **Standard deviation:** $s = \sqrt{s^2}$
- **Coefficient of variation (CV):** $CV = \frac{s}{\bar{X}}$
Measures relative variability

	A	B	C	D	E	F	G	H
1	Patient ID	Gender	Treatment Group	Age	Blood Pressure (mmHg)	Cholesterol (mg/dL)	Symptom Severity	Outcome
2	P001	Female	Control	34	120	190	Mild	Improved
3	P002	Male	Treatment A	45	138	220	Moderate	Stable
4	P003	Female	Treatment B	29	115	180	Mild	Improved
5	P004	Male	Control	54	142	245	Severe	Worsened
6	P005	Female	Treatment A	41	130	200	Moderate	Improved
7	P006	Male	Treatment B	38	128	210	Mild	Stable
8	P007	Female	Control	60	145	230	Severe	Worsened
9	P008	Male	Treatment A	50	135	215	Moderate	Stable
10	P009	Male	Treatment A	48	138	226	Moderate	Stable
11	P010	Female	Control	48	129	214	Mild	Worsened
12	P011	Male	Treatment A	68	139	194	Moderate	Improved
13	P012	Female	Treatment A	56	138	206	Moderate	Stable
14	P013	Female	Control	25	126	175	Severe	Worsened
15	P014	Female	Treatment B	45	141	233	Severe	Improved
16	P015	Male	Treatment A	66	140	222	Moderate	Stable

Dotplots



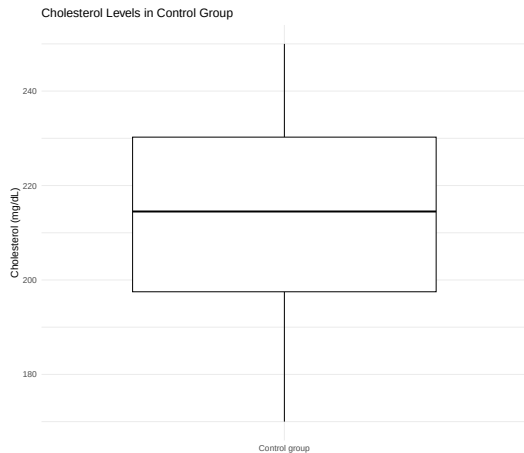
Dotplot Charts: Customizations



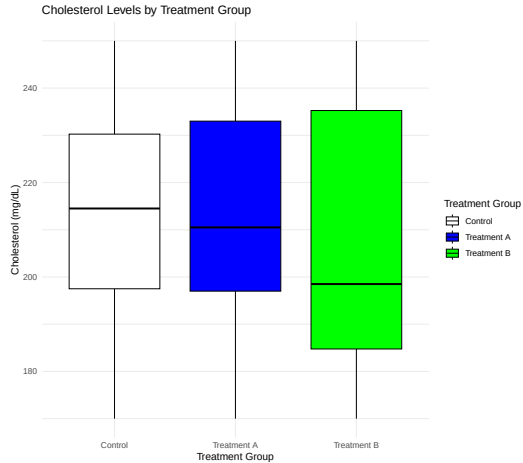
Boxplots

From summary statistics:

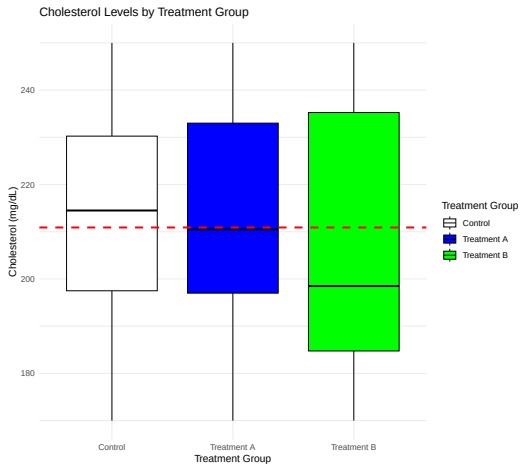
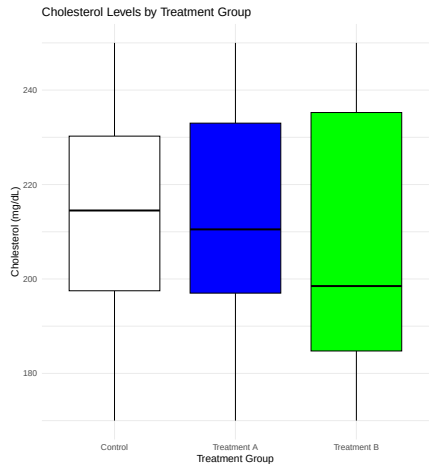
- $\text{Min} = 170$, $\text{Max} = 250$
- $\text{Mean} = 215.5$, $\text{Median} = 214.5$
- $Q_1 = 197.5$, $Q_3 = 230.2$



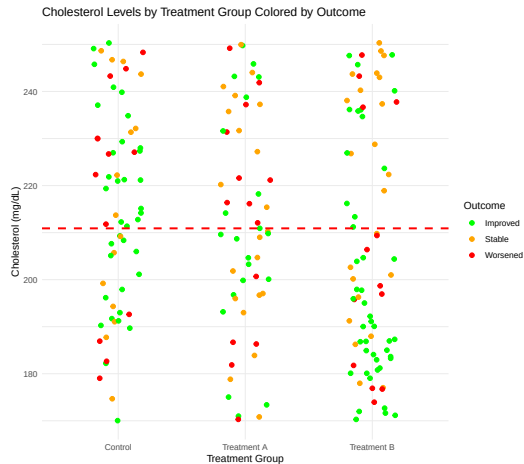
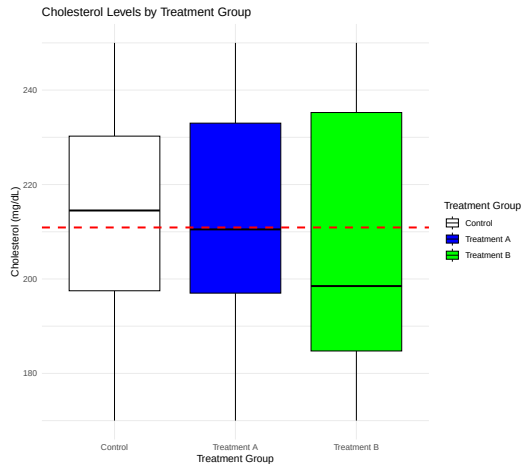
Boxplot Charts: Side-by-Side



Boxplot Charts: Side-by-Side

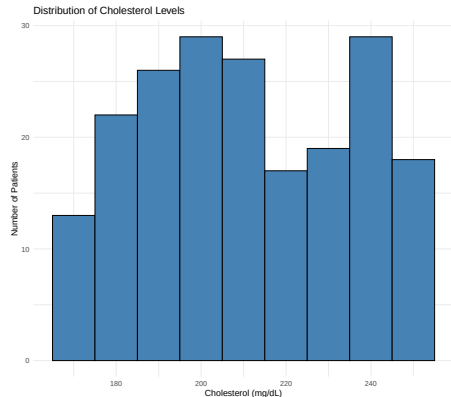


Boxplots vs. Dot Charts

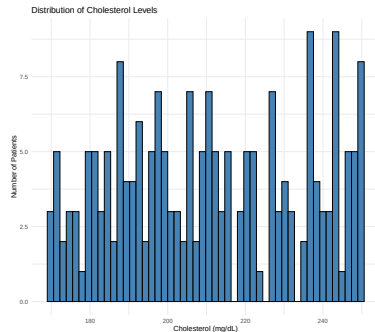
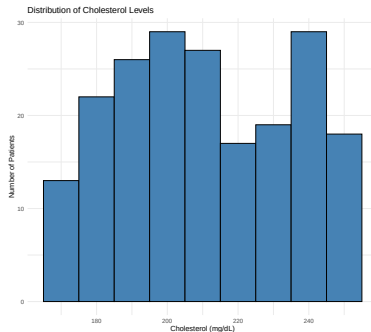
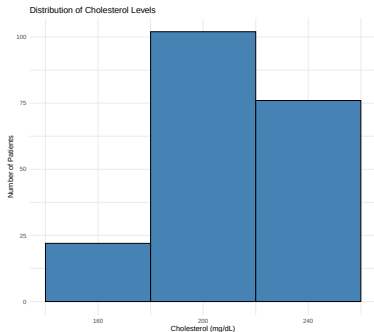


Histograms: Overview

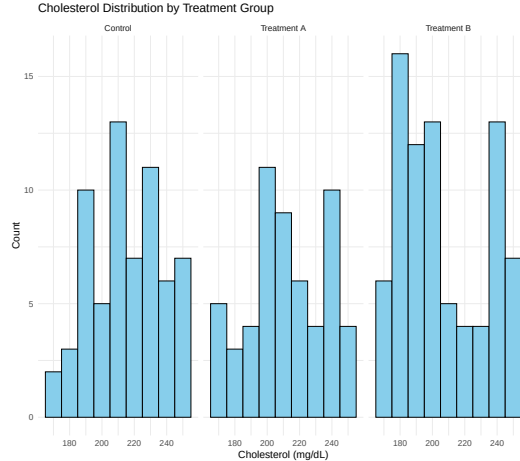
- Divide data into bins (intervals)
- Count frequency of data in each bin



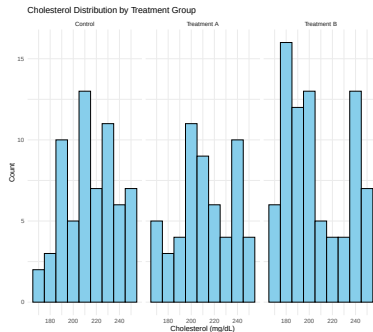
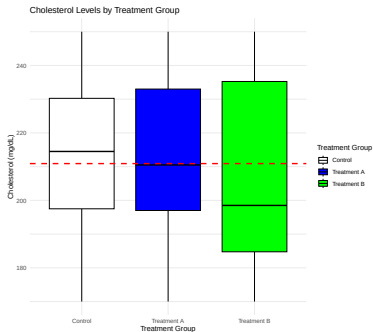
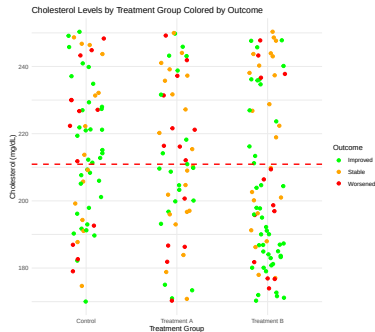
Histograms: Choice of Bins



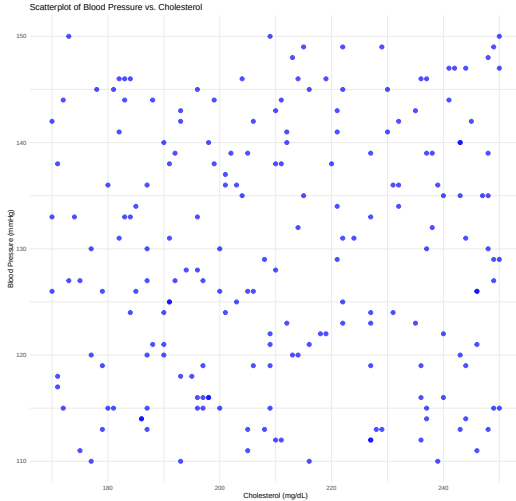
Side-by-Side Histograms



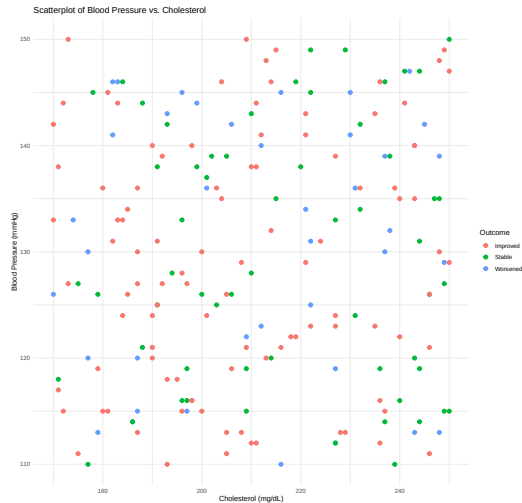
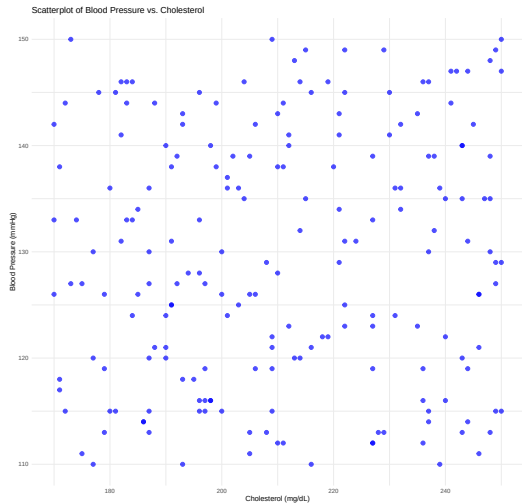
Summary: Graphical Displays of Numerical Data



The Scatterplot



The Scatterplot



Key Takeaways

- **Numerical summaries** describe center (mean, median) and spread (standard deviation, variance, CV)
- **Dotplots** and **boxplots** show distribution and grouping well; boxplots summarize five-number summary
- **Histograms** show overall distribution shape and are sensitive to bin width
- **Scatterplots** visualize relationships between two quantitative variables
- **Use multiple plots together** for deeper insight (e.g., dotplot + boxplot + histogram)
- Choose the right tool for your goal: description, comparison, or detecting patterns